

COMS20006

Software Engineering Project
Introduction to Cloud Computing

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What is Cloud Computing?

Cloud computing is the on-demand delivery of IT resources via the internet, replacing the need for traditional bare-metal hosting.

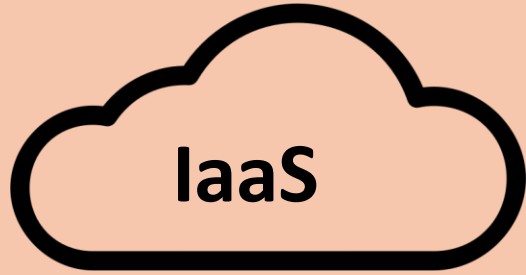
In previous years of SEP, we have used several other cloud providers, such as Azure, IBM Cloud and AWS Academy. However, we have found that these providers are somewhat restrictive in what they offer.

This year, you will be using Amazon Web Services (AWS). This should offer you full freedom and flexibility in comparison to the other providers.



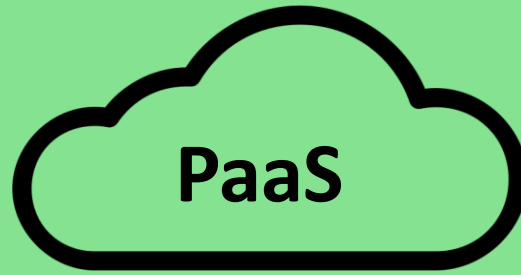
Service Models

AWS offers three primary service models:



Infrastructure as a Service

Provides virtualized computing resources, such as servers and storage, over the internet (e.g., AWS EC2 for scalable virtual machines).



Platform as a Service

Offers a fully managed environment for developing and running applications without managing infrastructure, allowing developers to run code in response to events without provisioning servers (e.g., AWS Lambda for serverless computing).



Software as a Service

Delivers complete software solutions accessible over the internet, with no need for user-managed infrastructure (e.g., Amazon Chime for video conferencing and collaboration).

Interacting with AWS

AWS Management Console

Web-based interface for accessing and managing AWS services visually. I'd highly recommend using the management console because it provides significant ease of use.

AWS CLI (Command Line Interface)

Command line tool that allows users to interact directly with AWS services. Useful for automation and scriptable tasks. For more information, you can find the documentation online.

AWS SDK (Software Development Kit)

AWS provides several SDKs for various programming languages such as Python (Boto3), JavaScript, and Java. SDKs are useful for programmatic access to AWS from within your application.

IaaS (Infrastructure as a Code)

IaaS allows you to define and manage AWS infrastructure through code. I'd recommend Terraform or AWS CloudFormation if you decide this is the best option.

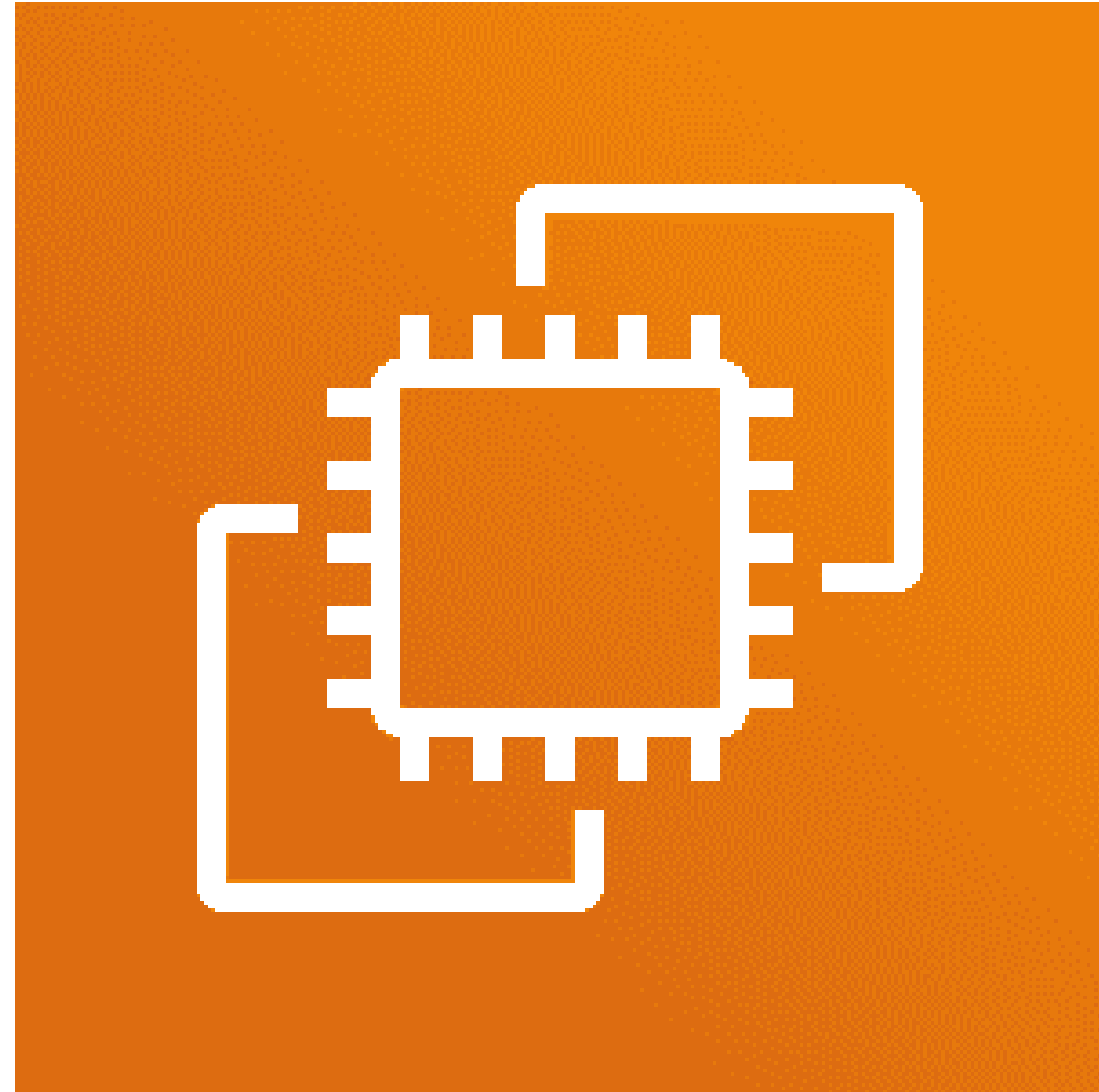
AWS Compute Services – EC2

AWS EC2 provides virtual machines. They are designed to be flexible, scalable cloud-based computing solutions with on-demand pricing.

The key idea is that EC2 allows users to scale compute resources up or down based on demand. Users can choose from a variety of instance types tailored for specific tasks, including general-purpose, compute-optimized, memory-optimized, and storage-optimized instances.

Key Benefits:

- **Cost Efficiency:** Pay only for the compute time you use.
- **Scalability:** Easily adjust the number of instances based on traffic patterns.
- **Customization:** Choose the operating system, instance type, and configurations to meet specific application needs.



AWS Compute Services – ECS & EKS

AWS Elastic Container Service (ECS) is a fully managed container orchestration service that allows you to deploy, manage and scale containerised applications. Very useful for teams that are bundling their application using something like Docker.

ECS provides the following functionality:

- Flexible Deployment Options: ECS offers both ECS on Fargate and ECS on EC2, providing options for both serverless compute and full instance control.
- Load Balancing: Integration with Application Load Balancers (ALBs) and Network Load Balancers (NLBs). Very useful for deploying both a backend and frontend and deciding how to route traffic across containers.

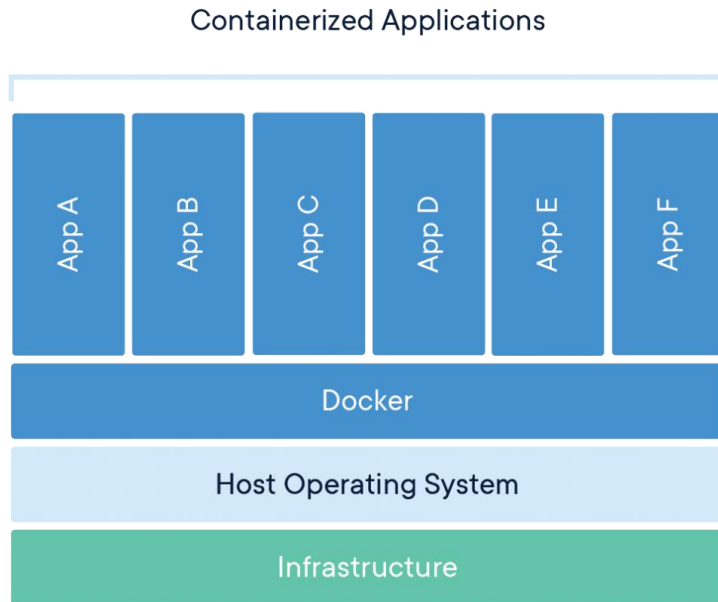
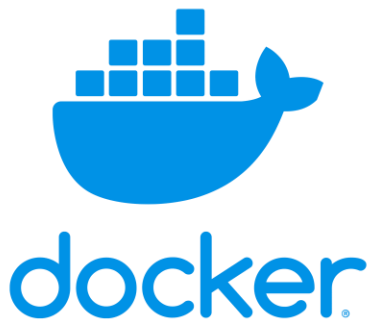
AWS Elastic Kubernetes Service (EKS) offers a similar service but instead manages the orchestration of Kubernetes clusters. This is probably out of the scope of SEP as it's a lot more flexible and complex and exceeds the needs of most projects.



Docker and Kubernetes

Docker is an extremely powerful open-source platform that enables developers to automate the deployment of applications in lightweight, portable containers.

- Docker ensures consistency across environments.
- Docker is highly scalable.



Dockerfiles provide the blueprint for building a Docker image. They define the base image, environment variables, file copies, and commands to run when the container starts.

Dockerfiles can be built using ``docker build`` to create a Docker image. A **Docker image** is a lightweight, stand-alone, and executable package that contains everything needed to run a piece of software, including application code, dependencies, configuration files and the runtime environment.

Docker images can be stored remotely on the cloud. They can then be pulled onto a machine and instantiated using ``docker run``. This will create a container. A **container** encapsulates a Docker image, utilising operating system-level virtualisation to isolate applications from one another.

AWS Compute Services – Lambda

Serverless compute service that allows you to execute code without provisioning or managing servers or virtual machines.

Can automatically scale based on the number of requests.

Convenient solution for building and deploying microservices, automation and event-driven applications.

Event-Driven Architecture

Lambda functions are triggered by events from AWS services, such as S3, DynamoDB, or HTTP requests through API Gateway.

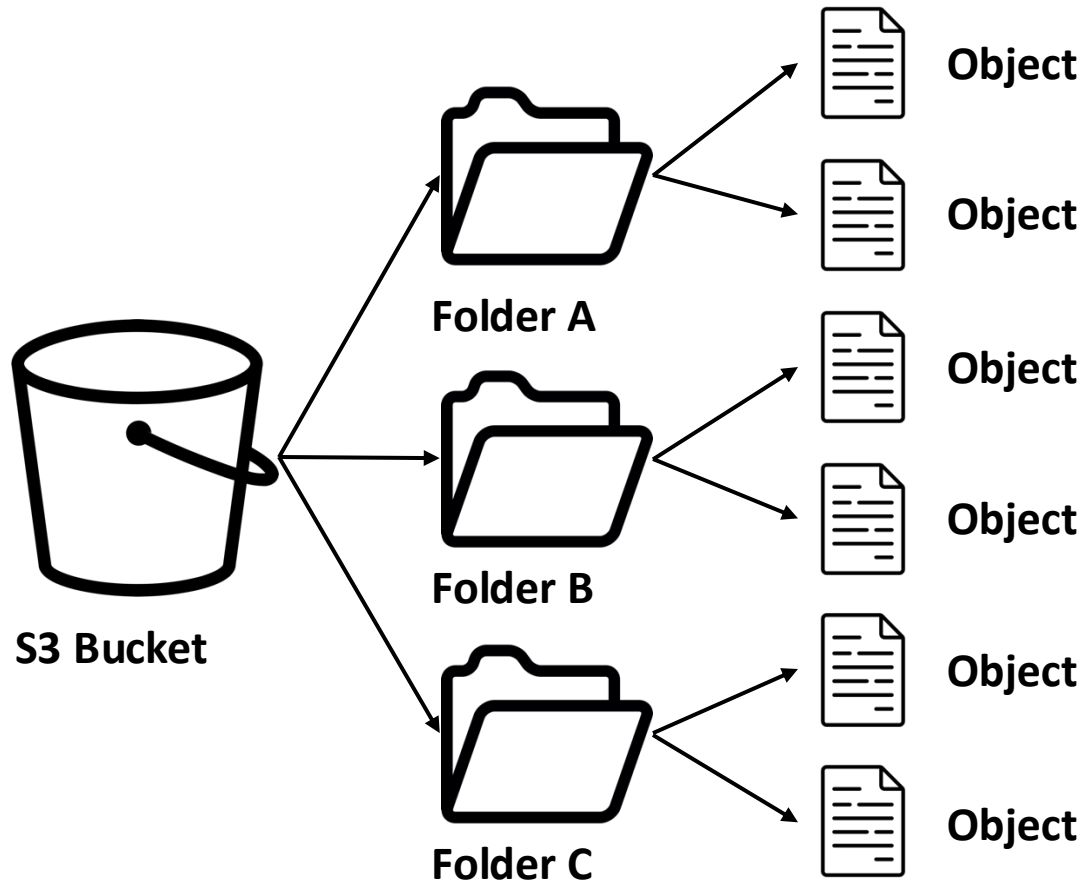


AWS Lambda natively supports several languages:

- Java
- Go
- PowerShell
- NodeJS
- C#
- Python
- Ruby

AWS Storage Services – S3

AWS S3 provides a scalable, secure object storage service designed to store and retrieve any amount of data from anywhere. All data is stored as objects or folders in buckets, with each object having a unique key.



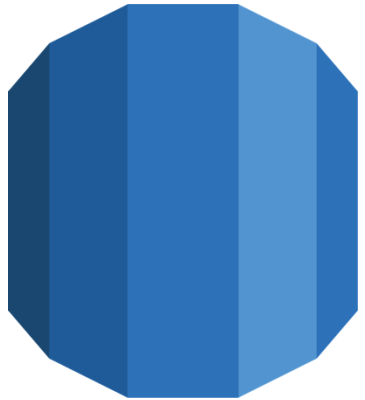
S3 is great for serving static content that is used within your project.

Some examples:

- Static websites built with HTML, CSS, and JavaScript.
- Image files (PNG, JPEG and GIF)
- Video files (MP4 and WebM)
- Documents and PDFs

AWS Storage Services – RDS & DyanmoDB

AWS provides a wide variety of fully managed database services designed to meet diverse application requirements, from SQL to NoSQL databases to specialised and purpose-built solutions. Some examples of services that could be used for your project:



Amazon RDS (Relational Database Service)

- SQL
- Managed relational database service
- Supports popular engines such as MySQL and PostgreSQL
- Can be backed up using S3
- Scales on demand



Amazon DyanmoDB

- NoSQL
- Serverless, key-value and document-based
- Quick setup
- Scales extremely well based on demand

AWS Networking Service – VPC & Route 53



Route 53

A scalable Domain Name System (DNS) service that translates domain names into IP addresses for seamless internet routing.

Very useful for setting up custom domain names within AWS, including routing traffic between different services.



VPC

A service that allows users to define isolated networks within the AWS cloud, providing full control over networking configurations and security.

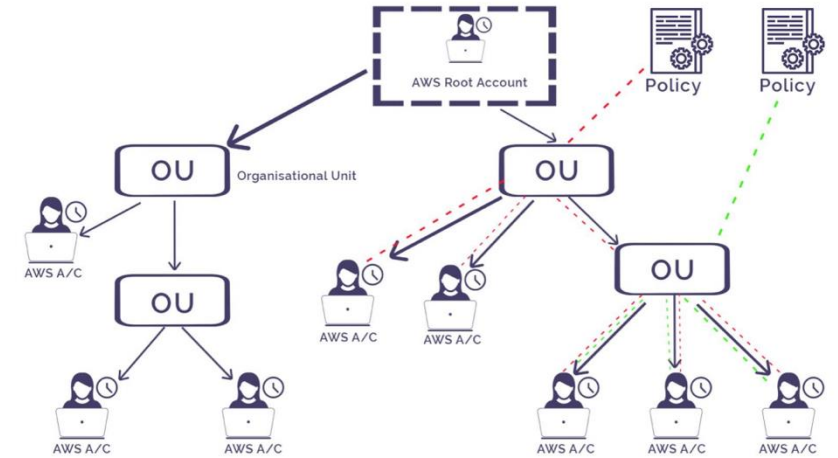
Probably will not be that useful within SEP but is an important part of cloud architecture.

AWS Identity and Access Management (IAM)

You will have almost complete control *over* your **Account**. Every SEP account belongs to the SEP Organisation Unit which will be managed by the AWS administrator.

When accounts are allocated, a budget will be enforced on your account, which will be strictly monitored.

Exceeding this budget will not immediately delete your account, however, restrictions will be placed on your services. You must contact Sarah or Jed if you exceed your budget as soon as possible.



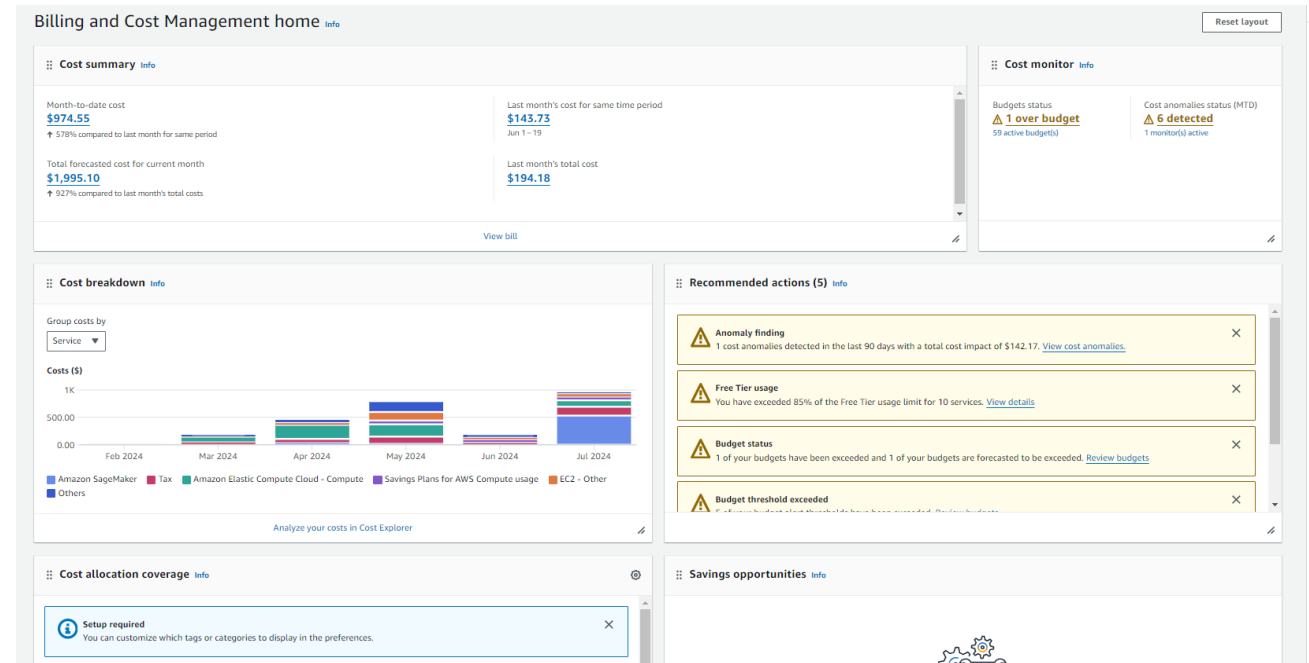
When your account is provisioned, a nominated member of your team will be provided root user credentials to the account. It is the nominated **User's** responsibility to set up credentials for the team. You must also provide the Unit Director, Jed Priest and your mentor with credentials.

AWS Pricing

When using AWS, pay attention to the costs of deploying services. Different services have different costs. You should **NEVER** add your own credit/debit card. Some top tips:

- Try not to deploy services in multiple Availability Zones.
- Use smaller instances for development.
- Turn off resources you aren't using.

AWS Admin Dashboard



Please remember that we are working with real money, the services that you use are billed as you use them. You need to be responsible with your usage. You ARE being monitored by university admins.

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A quick message from the Head TAs :)